INFORMATION TECHNOLOGY

Objectives

- In this session, you will learn to:
  - Solve programming problems by using stacks

Solving Programming Problems by Using Stacks

- Consider an example of a card game called Rummy.
- To start the game, a stock pile and a discard pile are placed in the center.
- A player can draw either the topmost card of the stock pile or the topmost card of the discard pile.
- If the drawn card does not make a valid sequence in the player's hand, the player can discard the card by placing it on the top of the discard pile.
- To represent such type of a discard pile in a computer program, you need a data structure that allows insertion and deletion at only one end.
- It should ensure that the last item inserted is the first one to be removed.

Defining a Stack

- A data structure that implements this concept is called a stack.
- A stack is a collection of data items that can be accessed at only one end, called top.
- Items can be inserted and deleted in a stack only at the top.
- The last item inserted in a stack is the first one to be deleted.
- Therefore, a stack is called a Last-In-First-Out (LIFO) data structure.

Data Structures and Algorithms

Identifying the Operations on Stacks

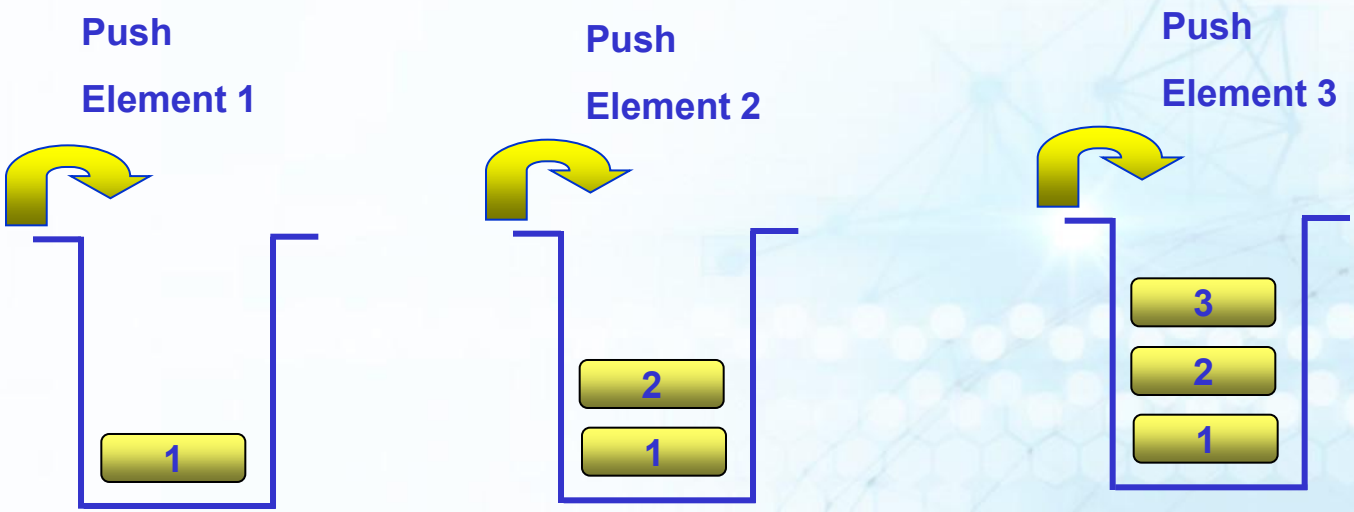
- There are two basic operations that are performed on stacks:
  - PUSH
  - POP

Empty Stack

Data Structures and Algorithms

Identifying the Operations on Stacks (Contd.)

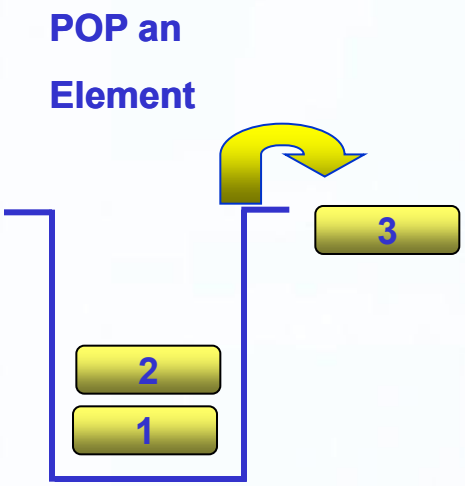
- PUSH: It is the process of inserting a new element on the top of a stack.

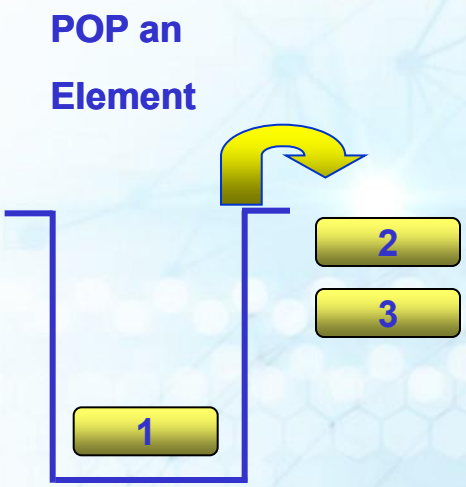


Data Structures and Algorithms

Identifying the Operations on Stacks (Contd.)

- POP: It is the process of deleting an element from the top of a stack.





Just a Minute

- Elements in stacks are inserted and deleted on a _____ basis.

Just a Minute (Contd.)

- Solution:
  - LIFO

Implementing a Stack

- Representing a stack in a linked list:

```
// Code in Java:
class Node
    {
        public int info;
        public Node next;
        public Node(int i, Node n)
        {
            info = i;
            next = n;
        }
    }
```

Data Structures and Algorithms

Implementing a Stack (Contd.)

```
class Stack
    {
        Node top;
        public Stack()
        {
            top = null;
        }
        bool empty() /* Returns true if the stack is
                        empty, false otherwise*/
        {            // Statements        }
        public void push(int element)
        {            // Statements        }
        public void pop()
        {             // Statements        }
}
```

Data Structures and Algorithms

Implementing a Stack (Contd.)

```
// Code in C++:

class Node
{
     public:
     int info;
     Node *next;
     public:
     Node(int i, Node* n)
        {
            info = i;
            next = n;
        }

};
```

Data Structures and Algorithms

Implementing a Stack (Contd.)

```
class Stack
{
        private:
        Node *top;
        public:
                Stack()
                {
                        top = NULL;
                }
                int empty()
                {       // Statements }
                }
                void push( int element)
                {       // Statements  }
                int pop()
                {       // Statements   }
};
```

Implementing a Stack (Contd.)

- The algorithm for PUSH operation by using a linked list is:

```
1. Allocate memory for the new node.
2. Assign value to the data field of the new node.
3. Make the next field of the new node point to top.
4. Make top point to the new node.
```

Implementing a Stack (Contd.)

- The algorithm for POP operation by using a linked list is:

```
1. If top = NULL:
    a. Display "Stack Underflow: Cannot delete from an
       empty stack".
    b. Exit.
2. Make tmp point to the topmost node.
3. Retrieve the value contained in the topmost node.
4. Make top point to the next node in sequence.
5. Release the memory allocated to the node marked by
   tmp.
```

Just a Minute

- In a stack, data can be stored and removed only from one end of the stack called the _____ of the stack.

Just a Minute (Contd.)

- Solution:
  - Top

Just a Minute (Contd.)

- If a stack is represented by using a linked list, then insertion and deletion of data will be done at the _____ of the list.

Just a Minute (Contd.)

- Solution:
  - Beginning

Demo: Solving Programming Problems by Using Stacks

- Problem Statement:
  - Write a program to implement a stack by using a linked list.

Demo: Solving Programming Problems by Using Stacks (Contd.)

- Solution:
  - To implement a stack by using a linked list, you need to perform the following tasks:
    1. Write the code.
    2. Execute the code and verify the output.

Summary

- In this session, you learned that:
  - A stack is a collection of data items that can be accessed at only one end, which is called top.
  - The last item inserted in a stack is the first one to be deleted.
  - A stack is called a LIFO data structure.
  - The following two basic operations can be performed on stacks:
    - PUSH
    - POP
  - A stack can be implemented by using an array or a linked list.